A

SYNOPSIS REPORT

On

AI-POWERED SMART TRAFFIC DIGITAL TWIN SYSTEM

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Topic of the Project: AI-Powered Smart Traffic Digital Twin System - A Role-Based Web Application for Adaptive Congestion Optimization and Priority Routing

1. Introduction

Modern urban centers face severe challenges due to rapid vehicle growth, resulting in chronic traffic congestion, increased carbon emissions, and delayed emergency response times. Conventional static traffic signal systems operate on pre-determined timing schedules that fail to adapt to live variations. The AI-Powered Smart Traffic Digital Twin System is a full-stack, role-based web application designed to create a real-time virtual replica of a city's traffic grid, enabling autonomous signal optimization, commuter delay forecasts, and emergency path clearing. The platform is built using React.js on the frontend and a high-performance Python FastAPI backend communicating with a cloud MongoDB Atlas instance.

The system provides two distinct portals - an Administrator (Traffic Police) portal and a Citizen portal - each with a custom dashboard. Administrators can monitor real-time node diagnostics, view simulated camera feeds, manually override traffic light phases (Red/Yellow/Green timers) under emergencies, and manage E-Challan records. Citizens, in turn, can check active travel advisories, view delay forecasts and weather-adjusted speed limits, search for registered vehicle challans, and pay them via digital payment integrations. By replacing legacy controls with a digitized visual digital twin, the system improves transit flow and coordinates city grids sustainably.

2. Literature Review

A total of eight research works published after 2020, drawn from IEEE and other reputed international publishers, were reviewed to identify existing approaches to traffic signal controls, emergency routing, and digital twin architectures. Several works focus on YOLO-based visual classification for vehicle detection [1]-[3], while others analyze dynamic routing shortcuts based on static Dijkstra metrics [4],[5]. Separately, web-based RTO systems for tolling and challans have matured independently of signal timing research [6],[7]. Across this body of work, a consistent gap emerges: real-time camera signal calculations, emergency green-wave dispatches, and public challan registries are treated as separate problems solved by single-purpose systems. No reviewed work integrates all three into a single accessible, role-based web platform - which is the specific gap this project fills.

3. Objective and Scope

The primary objective of the Smart Traffic Digital Twin System is to provide a single, centralized web-based portal through which traffic administrators can manage their grid nodes, and citizens can obtain live transit warnings. Specific objectives include:

- To provide secure, role-based login and registration for Admins and Citizens.
- To map real-world intersections coordinates using Leaflet and render live traffic congestion status.
- To automate green wave routing (Dijkstra algorithm) and prioritize emergency dispatches.
- To estimate real-time carbon offsets (CO_2 Saved index) and fuel savings based on vehicle counts.
- To enable vehicle license plate lookups and digital E-Challan payments via Razorpay.
- To provide safety speed advisories adjusted for weather limits and active police overrides.

The scope of the current phase is limited to a web-based twin prototype; hardware edge camera deployments with microcontrollers are identified as subsequent future scope items.

Mapping to Sustainable Development Goals (SDGs):

- SDG 3 - Good Health and Well-Being: Faster dissemination of emergency corridors reduces dispatch delays.
- SDG 9 - Industry, Innovation and Infrastructure: Digitizing signal controllers modernizes urban networks.
- SDG 11 - Sustainable Cities and Communities: AI-optimized phases reduce fuel idling and CO_2 emissions.

4. Hardware Requirements

The project does not require specialized hardware for its current simulation phase:
- Development Machine: Laptop/PC with a minimum of 8 GB RAM and multi-core CPU.
- Testing Devices: Responsive layout validation on standard mobile/desktop browsers.
- (Future scope): Edge AI hardware modules (NVIDIA Jetson) and IP CCTV cameras.

5. Software Requirements

- Frontend: React.js, Leaflet.js, HTML5, CSS3, ES6+.
- Backend: FastAPI (Python), PyMongo, Uvicorn Server.
- Database: MongoDB Atlas (Cloud database).
- Development: VS Code, Postman (API testing), Git & GitHub.

6. Methodology / Planning of Work

The project follows an incremental agile methodology, progressing through these stages:
- Requirement Analysis: Outlining the distinct views and roles of traffic admins and citizens.
- Database modeling: Constructing collections for intersections, challans, and dispatches.
- API Orchestration: Creating FastAPI endpoints for live telemetry queries and Dijkstra routing.
- Interface Design: Designing glassmorphic dashboards for signals, reports, and vehicle registries.
- Testing & Calibration: Calibrating GPS coordinates and offline fallback scenarios.

7. Testing Technologies

- Unit testing: Validation of FastAPI route handlers and route logic using pytest.
- API Verification: Swagger UI and Postman collections.
- User Acceptance Testing (UAT): Walkthroughs of citizen maps and admin signal override tools.

8. Resources and Limitations

- Resources: Open-source web modules (Vite, Leaflet, FastAPI) and free Nominatim geocoding services.
- Limitations: Requires browser coordinates permission; if denied, resolves IP fallback estimations.

9. Expected Outcome

The expected outcome is a fully functional web-based Smart Traffic Digital Twin system that maps grid nodes interactively, adjusts timers dynamically, clears paths for ambulance dispatches, estimates environmental offsets, and handles challan fine payments cleanly.

10. References [IEEE Format]

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